A Multimodal Sensor–HAR–Language Architecture for Goal-Aware User Interaction in Edge–Fog Smart Environments

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Abstract:
Context-aware assistance in a shared home depends on recognising the interlocutor and distinguishing observations, inferences and user declarations. We present a four-layer sensor–HAR–language reference architecture with explicit contracts for evidence, temporal scope and reflexive validation. A household case study uses Reachy Mini for face recognition and three local language-model roles to resolve metrics, select context and compose replies. A four-resident trial reported 96% target-user recognition over 50 attempts. In a 50-turn development replay across 15 scenarios, revised prompts and more selective evidence increased evidence-consistent replies from 34 to 41 and correct numerical answers from 7/8 to 8/8. In a separate 12-pair control, the model recovered every requested detail when supplied as an event. When the event was withheld, ten replies acknowledged the missing detail and two gave unsupported answers. The evaluation identifies remaining failures in source attribution and conversational continuity.

1 INTRODUCTION

Smart environments combine ambient sensors, wearables and computational services to analyse daily activities (Cook and Das, 2007). Edge–fog architectures distribute processing near acquisition, with connectivity, latency and privacy influencing the deployment (Shi et al., 2016). Social robots add an embodied interface, while user acceptance remains important in assistive applications (Broadbent et al., 2009).

Human activity recognition (HAR) connects measurements to interpretable descriptions through acquisition, segmentation and classification (Bulling et al., 2014). Data-driven methods learn patterns from multimodal streams (Ordóñez and Roggen, 2016). Knowledge-driven approaches use rules, ontologies or other explicit relationships (Polo-Rodríguez et al., 2025b). In shared homes, UWB localisation and proximity can help relate events to users (Polo-Rodríguez et al., 2024b; Polo-Rodríguez et al., 2024a). Identity at the robot, wearable identity and room location nevertheless refer to different observations.

Connecting this context to a large language model (LLM) raises two questions. How can the robot identify its interlocutor while preserving the provenance of the other modalities? How can a dialogue retain that provenance when the person asks about an earlier activity or corrects a sensor interpretation? A reply can contain an accurate reading yet overstate what it establishes. In this work, goal awareness means using a preference that the person states, such as moving more or adding vegetables, to orient an optional suggestion in the current conversation.

We contribute a four-layer sensor–HAR–language architecture with a distinct visual session-selection path, a local prototype that exchanges time-bounded metrics and source-tagged events across three prompted roles, and a household feasibility assessment that reports facial recognition, response consistency and event recovery separately.

2 RELATED WORK

We compare representative designs by how they acquire context, how they expose it to an application, and what they evaluate. The comparison concerns architectural choices rather than a common performance benchmark.

2.1 Smart-Home Middleware, Sensing and Robotics

The Context Toolkit separates acquisition and representation from application use (Dey et al., 2001). The Gator Tech Smart House and HomeOS shield applications from heterogeneous devices through service and platform abstractions (Helal et al., 2005; Dixon et al., 2012). The universAAL extension by Zmuda et al. (Zmuda et al., 2016) adds explicit semantics to assistive services. Our architecture shares the separation of device integration and application logic. The prototype uses Home Assistant, MQTT and a small typed query interface. This keeps the dialogue independent of device protocols, while dynamic discovery and a reusable semantic service platform remain outside the evaluation.

CASAS illustrates how timestamped residential events support longitudinal sensing and activity analysis (Cook et al., 2013). SPHERE similarly brings environmental, wearable and video observations together in a residential infrastructure (Elsts et al., 2018). Our sensor archive serves a similar role, but the Reachy Mini camera is used to select the interlocutor's session. Activity hypotheses come from wearable and ambient observations. The resulting evaluation distinguishes a visual assignment error from an episode or dialogue error.

Robot-Era connects assistive robots to ambient intelligence and evaluates service acceptability with older adults (Cavallo et al., 2018). CloudIA studies personalised robot-mediated activities in nursing facilities (Sorrentino et al., 2023). Both show the value of evaluating embodiment and interaction with users. Our study reports household face recognition and text grounding as separate feasibility tests, leaving acceptance and integrated spoken interaction for a prospective evaluation.

Learned face aggregation can combine several frames into a template and adapt their weights to image quality (Yang et al., 2017). The Reachy Mini pipeline also uses multiple frames, with explicit crop ranking, per-frame distances and a median-and-agreement rule. This choice leaves an inspectable explanation for a rejected batch. Learned aggregation is a separate candidate for comparison on held-out camera trials.

2.2 LLM Agent Architectures for Smart Environments

Polo-Rodríguez et al. (Polo-Rodríguez et al., 2025a) provide a direct antecedent by connecting UWB, ambient sensing and HAR to a chatbot. We retain this sensing-to-language path and add a visual session interface, a turn-specific cutoff and source-tagged declarations. He and Jazizadeh (He and Jazizadeh, 2025) also route historical queries through tools. Their agent supports broader analyses through generated code, while our fixed catalogue makes each operation and its temporal scope inspectable.

Harmony. Harmony separates event interpretation, decision-making and device control around a local language model (Yin et al., 2025). It combines virtual device tasks, an 18-participant questionnaire and an exploratory five-participant interaction study. Its feasibility checks are a precedent for reflexive validation. We apply a similar separation to historical queries and evidence selection. The open question in our setting is whether the reply preserves the source, person and time attached to a measurement or estimate.

ContextAgent. ContextAgent combines egocentric video and audio with persona context to decide when assistance is warranted and which tools to call (Yang et al., 2025). Its evaluation uses constructed scenarios and a subset paired with verified raw sensory inputs. Separating context extraction from the decision to intervene is relevant to our design. Its metrics concern proactive decisions and tool calls. Our replay measures how well a reply uses an available measurement, activity estimate or declaration.

ProAgent. ProAgent adds tiered perception that uses low-cost cues to request richer sensing when needed (Yang et al., 2026). The prototype provides evidence about wearable operation and timing, including participant-annotated intervention windows and ratings of responses generated from their recordings. Its sensing policy suggests an extension for our infrastructure layer. Our reference time bounds a query about recorded evi-

dence, while ProAgent's time windows evaluate when an intervention is appropriate.

Sensible Agent. Sensible Agent adapts the content and modality of AR suggestions and lets users confirm or modify them (Lee et al., 2025). Its preliminary study compares that interaction with voice-initiated assistance. Experimenter-triggered behaviour isolates the interaction design under staged conditions. It leaves autonomous triggering to further validation. The user confirmation step is relevant to our dialogue layer when an estimate or proposed action needs clarification.

These evaluations substantiate different parts of the architectures. Benchmark scores, recorded field data and controlled interaction studies provide complementary evidence. Proposed capabilities beyond the conditions tested remain hypotheses. In shared homes, sustained autonomous operation requires evaluating sensing, identity and interaction together over time. The present study also has that limit. It evaluates visual recognition and evidence-grounded dialogue separately and treats proactive timing as an open question.

Sasha maps high-level goals to relevant device actions and feedback (King et al., 2024). Our goal-aware scope concerns suggestions grounded in what the user states. Research on conversational coaching and just-in-time interventions provides examples of designs for assessing longer-term physical activity or dietary outcomes (Bickmore et al., 2013; Nahum-Shani et al., 2018). SayCan and ReAct motivate validating language-driven choices against available capabilities or observations (Ichter et al., 2023; Yao et al., 2023). Here that validation applies to metric requests and episode identifiers.

SensorLLM addresses an earlier boundary in the processing chain by aligning motion-sensor time series with language for activity recognition (Li et al., 2025). That design can learn representations directly from sensor sequences. Our case study exposes the output of a rule-based HAR service to the model. This keeps the evidence for a candidate inspectable and makes the dialogue interface independent of the HAR algorithm. It also limits the activity vocabulary to the operational predicates available in the engine.

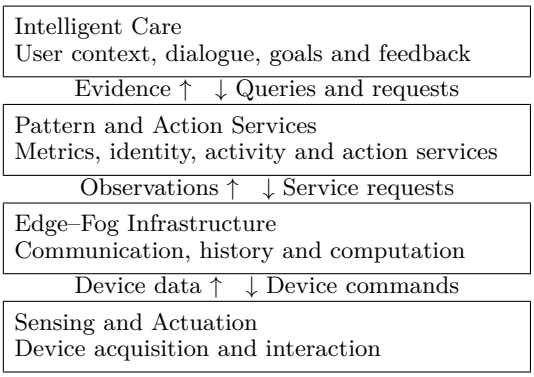


Figure 1: Reference architecture. Each layer exposes interfaces to the adjacent layers.

3 METHODOLOGY

3.1 Reference Architecture

We define a reference architecture for an intelligent environment that observes a shared space and interacts with its occupants. It separates acquisition, context interpretation and assistance. Figure 1 shows the four layers and the direction of their exchanges.

Sensing and Actuation. Sensors capture changes in the environment and signals related to occupants. Interaction devices acquire user input and deliver output. Each observation is associated with a source, a temporal reference and an entity or region when available. The device interface defines which actions it can execute and report back.

Edge–Fog Infrastructure. This layer connects devices and services. It manages message delivery, access to historical observations and placement of computation. Processing can be distributed near acquisition or across a local service infrastructure according to latency, resources and governance. The interfaces to the other layers are independent of that placement.

Pattern and Action Services. Pattern services derive quantities and contextual hypotheses from observations. Identity, localisation and activity recognition are separate services because they answer different questions. A knowledge-driven service may apply rules. A data-driven service may learn patterns. Both return evidence with the same scope and source information. Action services validate a request against the capabilities

Table 1: Evaluation scope of four LLM architectures for context-aware assistance.

Study	Architectural contribution	Evidence reported	Open question for residential dialogue
Harmony	Local modular reasoning with device feasibility checks	Virtual tasks, questionnaire and exploratory user interaction	Does a reply preserve the provenance and time of historical evidence?
ContextAgent	Sensory and persona context for proactive decisions and tools	1,000 benchmark samples, with a 300-sample sensory version	Do benchmark trigger decisions transfer to sustained use in shared homes?
ProAgent	Tiered on-demand sensing and timed assistance	92 h of recordings across 20 participants and ratings by those participants	How should personal sensing be combined with shared household records?
Sensible Agent	Context-sensitive AR suggestions and confirmation	Ten participants, 12 staged scenarios and experimenter-triggered behaviour	Can autonomous triggering preserve the observed interaction benefits?

and permissions of the deployment before execution.

Intelligent Care. The interaction layer uses evidence and user input to answer a question, explain an estimate or propose an action. A goal or preference can guide the response when relevant. Feedback from a user can revise the conversational interpretation. If an action is executed, its outcome can enter the sensing path as a new observation.

3.2 Evidence and Temporal Scope

An evidence item is represented as $e = (k, s, u, T, x)$, where k is the kind of evidence, s its source, u the subject or region, T a time point or interval, and x the payload. We distinguish an observation from an inference and a user declaration. An observation reports a reading or state. An inference carries a hypothesis and may include supporting observations or uncertainty. A declaration records what was communicated and when. If it describes an earlier period, the described interval is distinct from the time of the utterance.

An identity decision associates an interaction with a profile or returns unknown. This association has a scope, such as the current session. Linking another source to that profile requires an association of its own. A shared observation can remain unassigned even when the interaction identity is known.

A historical query is described by a target u, an operation o, an interval I and a reference time t_r. For a query about the past, $I = [a, b)$ with $a < b \leq t_r$. The response carries the result, the interval used and the evidence or coverage that qualifies it. A service may return partial evidence or an explicit unknown result. In a live system, the time at which an item became available can also constrain the query. An interval of supporting evidence and the duration of the phenomenon it suggests are distinct quantities.

3.3 Context Exchange and Extensibility

The dialogue process uses three functional roles. Metric resolution binds a request to an operation and temporal scope and obtains the result from a service. Semantic context selects relevant hypotheses, user declarations and events and preserves their sources. Assistive dialogue combines this context with the current user turn and applicable preferences. A deployment can combine roles in one component or implement them independently.

An event can become available between turns. The interaction layer may use it in a later response if it is relevant to the request. If the user corrects an inference, the correction can guide the conversation while the original observation remains in the history. Freshness and relevance are deployment policies. A record that is too old to describe the current state may still answer a question about the past.

Reflexive validation. Reliable interaction requires a way to check an interpretation before it is used in a reply or action. Given a request, selected evidence and a proposed result, a validation step can accept the result, request a revision or return unknown. For example, a check can verify that an answer refers to the requested interval and that a proposed command is supported by

Table 2: Recorded values per channel in the 4,320-minute replay.

Channel	Observed bins	Coverage
Steps	4,320	100.0%
Heart rate	3,768	87.2%
Watch bedtime mode	4,320	100.0%
Room estimate	2,529	58.5%
Television power	3,384	78.3%
Bathroom humidity	183	4.2%
Entrance sensor	81	1.9%

the environment. If a source or an association is missing, the system can ask for clarification or state what remains unknown. Harmony provides a precedent for combining context-grounded reasoning with feasibility checks before device execution (Yin et al., 2025).

An extension can replace a modality, a pattern service or the interaction interface by preserving its contract. For instance, an activity classifier can replace a rule engine if it exposes the hypothesis, its temporal scope and supporting evidence. A different identity service can select the interaction profile without changing the activity interface. The receiving component can therefore use the evidence type and scope without depending on the algorithm that produced it.

4 CASE STUDY

4.1 Sensor Setup and Record

Reachy Mini provides the camera, microphone and speaker. A Google Pixel Watch supplies steps, heart rate and its bedtime-mode flag. Home Assistant and MQTT integrate the observations locally (Froiz-Míguez et al., 2018). Quorvo UWB boards provide room estimates, while Aqara Zigbee and other Home Assistant channels record changes such as kitchen object use, humidity and electrical power. The fixed replay export contains 4,320 minute bins from 31 October 2026 at 12:06 to 3 November 2026 at 12:06, Europe/Madrid time, with an exclusive endpoint. Individual channels have different coverage (Table 2).

The preprocessed humidity and power features subtract the observed value from the median in a trailing window of two and 24 hours, respectively. They use the available portion of the window at the beginning of the record. A shower hypothesis separately compares a humidity observation to a known pre-bathroom reading and allows up to ten minutes after the last bathroom observation. A power event requires a rise of at least 20 W and a crossing to at least 20 W above the trailing median between observations no more than five minutes apart.

4.2 Historical Queries and Activity Rules

The Python engine indexes the fixed export by minute. For an interval with N expected bins and observed sets O_s and O_h, the step total is $\sum_{t \in O_s} s_t$, mean pulse is $\sum_{t \in O_h} h_t / |O_h|$ when available, and pulse coverage is $|O_h|/N$. A recorded zero is a measurement. Missing values remain unknown. Relative queries use the local calendar. The cutoff limits observation time in the fixed archive.

Table 3 lists the operational rules. The engine can bridge gaps of up to five minutes for most rules and retains the gap bins. It divides matching minutes by the duration of the candidate interval. The default threshold of 0.8 distinguishes supported episodes from weaker candidates. A candidate cut off by the query may represent only part of an activity.

4.3 Reachy Mini Camera and Face Recognition

The Reachy Mini client uses OpenCV and YuNet (Wu et al., 2023) to acquire a single complete face. Default acquisition requires a box of at least 120 pixels per side and an intersection over union of at least 0.65 with the first box. The client collects five distinct frames over at least one second. It ranks the face crops by sharpness, exposure and saturation, then sends the best three as JPEG crops to a local vision service.

The server aligns each face and obtains 512-dimensional FaceNet embeddings through DeepFace (Schroff et al., 2015; Serengil and Ozpinar, 2020). For frame i and enrolled profile u, it computes the minimum cosine distance from the crop embedding to the reference gallery G_u. A profile requires valid comparisons in at least two frames and is ranked by its median distance. Acceptance requires a best distance of at most 0.50, a margin of at least 0.05 to the next profile if available, and two individual frame matches. Insufficient or ambiguous evidence yields an unknown result. The visual decision can select a dialogue session, and the service supports bounded diagnostic capture storage. The reported evaluation uses aggregate outcomes without including face images or embeddings.

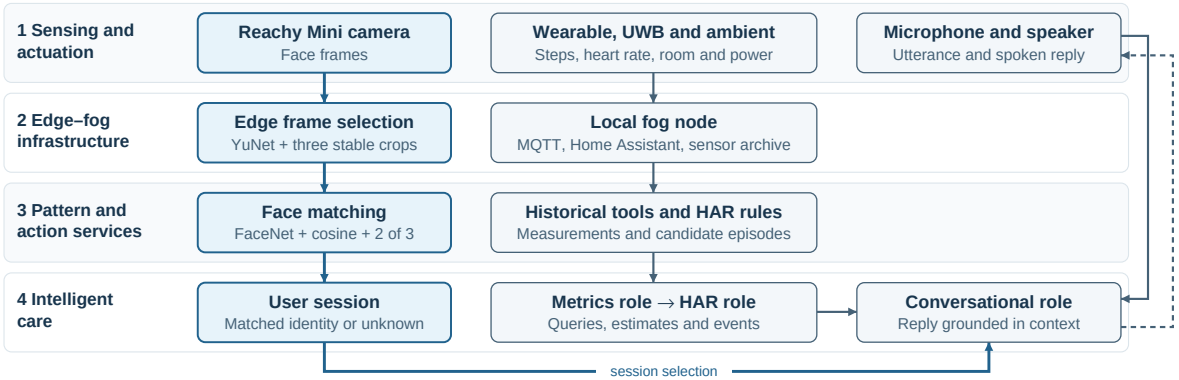


Figure 2: Household implementation of the reference architecture. The blue path connects face frames to session selection. The central path carries measurements and episode candidates. The dashed arrow returns the reply to the robot.

Table 3: Operational rules and interpretation of evidence in the household prototype.

Rule	Evidence	Interpretation
Exercise	At least 60 steps/min and 100 beats/min, with a candidate of at least five minutes	Compatible with ambulatory exercise. Summarise the observed interval.
Rest	Zero observed steps and at most 65 beats/min, over a candidate of at least ten minutes	Rest hypothesis with reported coverage.
Bathroom or shower	Bathroom room estimate. A shower candidate also requires an 8-point humidity rise over a known reference within the allowed delay.	Approximate shower hypothesis. Exact cause and duration remain unresolved.
Kitchen use	Kitchen room estimate plus fridge light of at least 10 lux or a glass or plate sensor event in the same minute	Kitchen-use cue, with a self-report preferred when describing what the user was doing.
Watch bedtime	Operating-system mode flag	Observed start, end or duration of the mode.
Entrance	Entrance-area sensor event	Direction and time away are unresolved.

The frame and batch decisions address different sources of error. Geometric stability controls acquisition, while the distance margin rejects similar profile matches. The median and two-frame agreement limit the influence of one inconsistent crop. The single-face acquisition condition also means that recognition can wait while several people share the camera view. The household trial does not report the frequency of these events, and the thresholds are configuration defaults rather than estimates fitted from the 50 attempts.

4.4 Local Agent Integration

The replay runs Qwen3-4B-Instruct-2507 in Q4_K_M quantisation through Ollama (Qwen Team, 2025). The coordinator uses three sequential calls per turn. The robot integration can select a resident session by face recognition. In the text replay the sensor identity is fixed and the scripts supply Spanish utterances at explicit times.

The Python coordinator validates the structured fields returned by the first two roles. Metrics selects an available operation from the user turn, reference time and recent dialogue. HAR Context selects an episode or abstains, and can extract a literal self-report. Conversation receives the selected evidence and a short transcript. Its instructions distinguish sensor readings, activity estimates and what the user said. These checks govern the query and evidence inputs. The visible reply is assessed by the offline audit. The implementation retains the latest event per source and the two preceding exchanges. The freshness windows are 10 minutes for power events and 15 minutes for user declarations.

In one scene, the user reports reading a novel in another room. While that topic continues, the television power changes from 1.0 to 70.6 W at 12:41. At the next turn both items are available.

The declaration is evidence about the user's activity and the power event is evidence about the appliance. The conversation can stay on the novel until the person asks about the television.

5 EVALUATION

We ask whether the camera can recognise the target user in the household (RQ1), whether dialogue replies remain consistent with supplied evidence (RQ2), and whether a structured event can carry a detail beyond the short transcript (RQ3). The three assessments use different units of analysis.

5.1 Household Recognition

The reported trial comprised 50 recognition attempts concerning the target resident in a four-person household, including her spouse and two children aged 11 and 14. Reported accuracy was 96%, equivalent to 48 correct outcomes. The available aggregate does not specify probe composition, error types or capture conditions. The result is target-user recognition within one household. A follow-up protocol can distinguish enrolled and non-enrolled probes and record false assignment, missed identification and abstention (Grother et al., 2019).

5.2 Dialogue Replay and Scoring

Fifteen scripted situations provide 50 user turns and 150 model calls. Scenes include kitchen use, shower estimation, exercise, the watch mode, rest, step totals, appliance events, user corrections and recall. Two scenarios introduce a physical activity or food preference. The query time can advance between turns to include new observations or test declaration age. The replay uses text inputs and one fixed sensor archive.

A single LLM-assisted rubric audit scores each visible reply. A pass requires the appropriate scope and source and an acknowledgement of uncertainty where evidence is insufficient. The numeric subset scores requested values independently of attribution, with a one-minute tolerance for an estimated shower time. The initial and revised configurations use the same 50 messages, query times, data and criteria. The revision makes the source and reference of selected readings explicit and gives the response model shorter instructions and relevant evidence for clear requests. For the last adjustment, dialogues were rerun from the first affected turn and unchanged prefixes were retained. This comparison describes development performance on these scenarios. Inference used temperature zero, a 4,096-token context and up to 160 output tokens per call.

Table 4 gives the scenario breakdown. In the exercise questions the engine returned a bounded result. In the humidity questions a correct number could still be accompanied by an incorrect explanation. The replay script records the original message, time cutoff and selected operation, together with the available episodes, role exchanges and final reply. This allows review at the level of a single answer. The trace identifies the preprocessed export by its SHA-256 digest. The event probe reuses the local model and fixtures from that archive, with scripted declarations supplied separately.

5.3 Paired Event-Payload Probe

The probe uses the initial dialogue configuration and invokes its conversational role twice per question. It holds t_0, model, prompt, short transcript, metrics and activity context fixed. The conditions differ only in whether the event is supplied, and their order alternates. Eight questions ask for a detail in a scripted self-report after two unrelated exchanges. Four ask for power immediately before and after a recorded jump. Both conditions retain the current power and its rolling-reference difference. A recall case passes when the target detail is present. A power case passes when both numbers appear in the correct order. When the event is removed, the target is unavailable by design. The control distinguishes appropriate abstention from an unsupported answer about that target.

5.4 Results and Failure Analysis

Evidence-consistent replies increased from 34/50 (68%) to 41/50 (82%). Eleven turns improved and four regressed. Sensor and metric grounding accounted for the largest gain, while context and event grounding fell from 19/22 to 18/22 and goal alignment from 3/3 to 2/3. The revised reply gave the correct watch-mode end, distinguished the time of a humidity reading from the start of a rise, and reported eight observed minutes, 858 steps and 109.5 beats/min for the exercise interval. It still missed a book detail present in the

Table 4: Scenario-level audit of the revised dialogue configuration. Each count gives replies meeting the turn-specific criterion over turns in that scenario.

Scenario	Pass / turns	Scenario	Pass / turns
Kitchen greeting and evidence	3/3	No reliable current activity	3/3
Recent shower estimate	1/3	Steps and temporal scope	3/3
Television and unknown viewer	3/3	Reading and delayed recall	4/6
Exercise and morning scope	3/3	Kitchen correction	3/4
Watch bedtime-mode end	3/3	Television event during dialogue	4/4
Rest and movement	2/3	Physical activity preference	2/3
Kitchen-use clarification	2/3	Food-choice preference	2/3
Humidity and activity	3/3	Total	41/50

Table 5: Recognition and paired dialogue replay. Numeric checks overlap with sensor and metric grounding.

Evaluation	Initial	Revised
Target-user face recognition	48/50 (96%)	
Dialogue evidence consistency	34/50	41/50
Sensor and metric grounding	12/25	21/25
Context and event grounding	19/22	18/22
Goal-aligned suggestions	3/3	2/3
Numeric subset	7/8	8/8

Table 6: Paired event-payload probe.

Requested detail	Event kept	Event removed
Prior self-report	8/8	0/8
Before/after power	4/4	0/4
Total	12/12	0/12

Table 7: Illustrative errors and changes between the two configurations.

Failure	Reply and evidence
Episode binding	The initial reply used a bathroom end at 09:52. The revision used the watch flag end at 09:57.
Reference semantics	The initial reply called the humidity reference a mean. The revision used the two-hour median.
Appliance claim	The revised reply inferred television state from the person's reading rather than using the measured 0.9 W.
User report	The revised reply missed a book genre that the user had stated and that the event payload retained.

event context. In the revised trace, median text-turn latency was 13.1 s and the 90th percentile was 18.9 s on an Intel Core i7-13620H CPU, excluding audio and camera acquisition.

The paired probe recovered 12/12 target details with the event and 0/12 with it removed (Table 6). One response recovered the detail while reversing who had said it. When the target was withheld, ten replies indicated that it was unavailable and two gave unsupported details. One named a different plant and one treated a current reading as the value before a power jump. This control exposes guessing when the required evidence is absent. The full dialogue replay separately tests selection and interpretation.

6 DISCUSSION

Typed metric queries make the calculation and time scope inspectable. They limit coverage to operations in the catalogue. A valid follow-up can require an interval or joint measure outside that catalogue. The watch-mode error illustrates how the correct source can be available while the model answers from the wrong episode. A future replay can separate follow-up resolution from composition and check that the answer uses the requested result.

Source tags also preserve distinctions without guaranteeing their use. The event probe isolates information recovery on deliberately constructed questions, with one model and one small set of inputs. It does not test extraction quality or compare the event list with a longer transcript or retrieval from an archive. The replay has correlated turns, development-time exposure and a single rubric review. Independent reviewers and held-out scenarios would provide stronger evidence of dialogue quality. Independently annotated activity and boundary times would be required to estimate HAR accuracy.

The visual result covers one household and an aggregate set of attempts. A prospective integrated study should record enrolled and unknown probes, acquisition conditions, rejection and wrong-session selections, along with spoken interaction outcomes. The present tests address visual recognition and text grounding separately. Local processing changes where data are handled, while consent, access and diagnostic retention depend on the deployment protocol.

Comparison with action-centred systems suggests testing device control separately. SimuHome (Seo et al., 2026) provides a way

to verify state changes and time-dependent workflows in simulation. A longer study of physical activity and food preferences would require outcomes and follow-up periods suited to those goals. Here, goal alignment describes the relevance of suggestions within the turn.

Reproducibility. The implementation separates the turn fixtures, model calls and review outcomes. The full replay runner generates the dialogue trace from the preprocessed export. The paired-probe runner accepts the dialogue trace and reconstructs the environmental event fixtures from the same export. It records both answers and the target-recovery decision for each pair. The time window and sensor snapshot must match when reproducing the numerical results. The rubric scores are tied to the recorded replies, so a fresh run should be reviewed against the same criteria. Facial recognition is evaluated separately from these scripts and would require a documented camera trial and access to its enrolment and test protocol.

7 CONCLUSIONS

The reference architecture separates observations, contextual services and interaction through reusable interfaces. Its household implementation connects Reachy Mini visual session selection with metric tools, HAR context and local dialogue. The study reported 96% target-user recognition, and evidence-consistent replies rose from 34/50 to 41/50 on the same development scenarios after revising prompts and evidence selection. The withheld-event control and the remaining response errors motivate checking what an answer asserts as well as which facts it retrieves.

REFERENCES

Bickmore, T. W., Schulman, D., and Sidner, C. (2013). Automated interventions for multiple health behaviors using conversational agents. Patient Education and Counseling, 92(2):142–148.

Broadbent, E., Stafford, R., and MacDonald, B. (2009). Acceptance of healthcare robots for the older population: Review and future directions. International Journal of Social Robotics, 1(4):319–330.

Bulling, A., Blanke, U., and Schiele, B. (2014). A tutorial on human activity recognition using body-worn inertial sensors. ACM Computing Surveys, 46(3):33:1–33:33.

Cavallo, F., Esposito, R., Limosani, R., Manzi, A., Bevilacqua, R., Felici, E., Di Nuovo, A., Cangelosi, A., Lattanzio, F., and Dario, P. (2018). Robotic services acceptance in smart environments with older adults: User satisfaction and acceptability study. Journal of Medical Internet Research, 20(9):e264.

Cook, D. J., Crandall, A. S., Thomas, B. L., and Krishnan, N. C. (2013). CASAS: A smart home in a box. Computer, 46(7):62–69.

Cook, D. J. and Das, S. K. (2007). How smart are our environments? an updated look at the state of the art. Pervasive and Mobile Computing, 3(2):53–73.

Dey, A. K., Abowd, G. D., and Salber, D. (2001). A conceptual framework and a toolkit for supporting the rapid prototyping of context-aware applications. Human–Computer Interaction, 16(2–4):97–166.

Dixon, C., Mahajan, R., Agarwal, S., Brush, A. J. B., Lee, B., Saroiu, S., and Bahl, V. (2012). An operating system for the home. In 9th USENIX Symposium on Networked Systems Design and Implementation (NSDI 12), pages 337–352.

Elsts, A., Fafoutis, X., Woznowski, P. R., Tonkin, E. L., Oikonomou, G., Piechocki, R., and Craddock, I. (2018). Enabling healthcare in smart homes: The SPHERE IoT network infrastructure. IEEE Communications Magazine, 56(12):164–170.

Froiz-Míguez, I., Fernández-Caramés, T. M., Fraga-Lamas, P., and Castedo, L. (2018). Design, implementation and practical evaluation of an IoT home automation system for fog computing applications based on MQTT and ZigBee-WiFi sensor nodes. Sensors, 18(8):2660.

Grother, P., Ngan, M., and Hanaoka, K. (2019). Face recognition vendor test (FRVT) part 2: Identification. Technical Report NISTIR 8271, National Institute of Standards and Technology.

He, T. and Jazizadeh, F. (2025). Context-aware LLM-based AI agents for human-centered energy management systems in smart buildings. arXiv preprint 2512.25055.

Helal, S., Mann, W., El-Zabadani, H., King, J., Kaddoura, Y., and Jansen, E. (2005). The Gator Tech Smart House: A programmable pervasive space. Computer, 38(3):50–60.

Ichter, B., Brohan, A., Chebotar, Y., Finn, C., Hausman, K., Herzog, A., Ho, D., Ibarz, J., Irpan, A., Jang, E., Julian, R., Kalashnikov, D., Levine, S., Lu, Y., Parada, C., Rao, K., Sermanet, P., Toshev, A. T., Vanhoucke, V., Xia, F., Xiao, T., Xu, P., Yan, M., Brown, N., Ahn, M., Cortes, O., Sievers, N., Tan, C., Xu, S., Reyes, D., Rettinghouse, J., Quiambao, J., Pastor, P., Luu, L., Lee, K.-H., Kuang, Y., Jesmonth, S., Joshi, N. J., Jeffrey, K., Ruano, R. J., Hsu, J., Gopalakrishnan, K., David, B., Zeng, A., and

Fu, C. K. (2023). Do as i can, not as i say: Grounding language in robotic affordances. In Proceedings of The 6th Conference on Robot Learning, volume 205 of Proceedings of Machine Learning Research, pages 287–318. PMLR.

King, E., Yu, H., Lee, S., and Julien, C. (2024). Sasha: Creative goal-oriented reasoning in smart homes with large language models. Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies, 8(1).

Lee, G., Xia, M., Numan, N., Qian, X., Li, D., Chen, Y., Kulshrestha, A., Chatterjee, I., Zhang, Y., Manocha, D., Kim, D., and Du, R. (2025). Sensible Agent: A framework for unobtrusive interaction with proactive AR agents. In Proceedings of the 38th Annual ACM Symposium on User Interface Software and Technology. ACM.

Li, Z., Deldari, S., Chen, L., Xue, H., and Salim, F. D. (2025). SensorLLM: Aligning large language models with motion sensors for human activity recognition. In Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing, pages 354–379, Suzhou, China. Association for Computational Linguistics.

Nahum-Shani, I., Smith, S. N., Spring, B. J., Collins, L. M., Witkiewitz, K., Tewari, A., and Murphy, S. A. (2018). Just-in-time adaptive interventions (JITAIs) in mobile health: Key components and design principles for ongoing health behavior support. Annals of Behavioral Medicine, 52(6):446–462.

Ordóñez, F. J. and Roggen, D. (2016). Deep convolutional and LSTM recurrent neural networks for multimodal wearable activity recognition. Sensors, 16(1):115.

Polo-Rodríguez, A., Anguita-Molina, M. Á., Gil, D., Romero, J., Fernández, E., Paloma, O., Porcel, A., and Medina-Quero, J. (2024a). Discovering social interactions between caregivers and frail individuals using indoor localization. In Proceedings of the International Conference on Ubiquitous Computing and Ambient Intelligence (UCAmI 2024), volume 1212 of Lecture Notes in Networks and Systems, pages 319–331, Cham. Springer.

Polo-Rodríguez, A., Cavallo, F., Nugent, C., and Medina-Quero, J. (2024b). Human activity mining in multi-occupancy contexts based on nearby interaction under a fuzzy approach. Internet of Things, 25:101018.

Polo-Rodríguez, A., Fiorini, L., Rovini, E., Cavallo, F., and Medina-Quero, J. (2025a). Enhancing smart environments with context-aware chatbots using large language models.

Polo-Rodríguez, A., López, I. V., Diaz, R., Rivadeneyra, A., Gil, D., and Medina-Quero, J. (2025b). Modelling key health indicators from sensor data using knowledge graphs and fuzzy logic. Electronics, 14(12):2459.

Qwen Team (2025). Qwen3-4B-Instruct-2507. Model card on Hugging Face.

Schroff, F., Kalenichenko, D., and Philbin, J. (2015). FaceNet: A unified embedding for face recognition and clustering. In IEEE Conference on Computer Vision and Pattern Recognition, pages 815–823.

Seo, G., Yang, J., Pyo, J., Kim, N., Lee, J., and Jo, Y. (2026). SimuHome: A temporal- and environment-aware benchmark for smart home LLM agents. In International Conference on Learning Representations.

Serengil, S. I. and Ozpinar, A. (2020). LightFace: A hybrid deep face recognition framework. In Innovations in Intelligent Systems and Applications Conference, pages 23–27.

Shi, W., Cao, J., Zhang, Q., Li, Y., and Xu, L. (2016). Edge computing: Vision and challenges. IEEE Internet of Things Journal, 3(5):637–646.

Sorrentino, A., Fiorini, L., La Viola, C., and Cavallo, F. (2023). Measuring the perception of the personalized activities with CloudIA robot.

Wu, W., Peng, H., and Yu, S. (2023). YuNet: A tiny millisecond-level face detector. Machine Intelligence Research, 20:656–665.

Yang, B., Xu, L., Zeng, L., Guo, Y., Jiang, S., Lu, W., Liu, K., Li, Y., Jiang, X., Xing, G., and Yan, Z. (2026). ProAgent: Harnessing on-demand sensory contexts for proactive LLM agent systems in the wild. arXiv:2512.06721, version 2. First posted in 2025. Accepted for publication in IMWUT.

Yang, B., Xu, L., Zeng, L., Liu, K., Jiang, S., Lu, W., Chen, H., Jiang, X., Xing, G., and Yan, Z. (2025). ContextAgent: Context-aware proactive LLM agents with open-world sensory perceptions. In Advances in Neural Information Processing Systems, volume 38.

Yang, J., Ren, P., Zhang, D., Chen, D., Wen, F., Li, H., and Hua, G. (2017). Neural aggregation network for video face recognition. In IEEE Conference on Computer Vision and Pattern Recognition, pages 4362–4371.

Yao, S., Zhao, J., Yu, D., Du, N., Shafran, I., Narasimhan, K., and Cao, Y. (2023). ReAct: Synergizing reasoning and acting in language models. In International Conference on Learning Representations.

Yin, Z., Zhang, M., and Kawahara, D. (2025). Harmony: A human-aware, responsive, modular assistant with a locally deployed large language model. In Companion of the 2025 ACM International Joint Conference on Pervasive and Ubiquitous Computing.

Zmuda, D., Psiuk, J., and Psiuk, M. (2016). Seamless semantic enrichment of services in assistive environments. Universal Access in the Information Society, 15:85–99.